

Ballot words and the Morin Thom polynomials

Chern++

September 2026

Abstract

The Chern coefficients $C(M)$ of the Morin Thom polynomials Tp_{A_d} are indexed by zero-sum multisets M of d integers. Let $b(M)$ be the number of orderings of M whose proper partial sums are nonnegative. We prove, given one theorem of an unreferenced external report, that $C(M) = b(M)$ whenever $\max M \leq 1$. On that sector $b(M)$ is a Kreweras number. We conjecture that $C(M) \geq b(M)$ for every M , with equality exactly on that sector. This implies Rimányi's positivity conjecture, and it holds, with its equality case, on every exact coefficient we can compute: all $d \leq 7$ on large boxes, including Tp_{A_7} to $\ell = 5$ and every packet of charge ≤ 6 at $d = 7$.

1 Setting

By Rimányi's ℓ -free reduction [2] and the Bérczi–Szenes residue formula [1],

$$\text{Tp}_{A_d}^\ell = \sum_M C(M) \prod_{a \in M} c_{\ell+1+a},$$

where M ranges over multisets of d integers summing to 0, $c_0 = 1$, and $c_j = 0$ for $j < 0$. The coefficient $C(M)$ does not depend on ℓ . In chamber coordinates,

$$C(M) = \sum_{\alpha} A_{\beta(\alpha)},$$

where the sum is over the distinct orderings α of M , the prefix sums $\beta_j = \alpha_1 + \dots + \alpha_j$ ($j < d$) must be nonnegative, and $F_d = \sum_{\beta} A_{\beta} x^{\beta}$ is the Morin chamber series. Rimányi's conjecture is $C(M) \geq 0$.

Definition 1. A *ballot ordering* of M is an ordering whose proper prefix sums are all nonnegative. Let $b(M)$ be the number of distinct ballot orderings. Equivalently, $b(M)$ is $C(M)$ computed with the pure chain series $\prod_{j=1}^{d-1} (1 - x_j)^{-1}$, all of whose coefficients are 1, in place of F_d .

The decreasing ordering is always ballot, so $b(M) \geq 1$. The *charge* of M is $\max M$. The *plane sector* is charge ≤ 1 ; for $\ell = L - 1$ these are the Chern monomials whose largest index is at most $\ell + 2$.

2 The plane sector

Theorem 2. Assume Theorem 5.4 of [3]. If $\max M \leq 1$ then $C(M) = b(M)$. Write $M = (1^{d-b}, 1 - s_1, \dots, 1 - s_b)$ with every $s_i \geq 1$, and let m_i be the number of s 's equal to i . Then this common value is the number of noncrossing partitions of $[d]$ with block sizes $s_1, \dots, s_b$:

$$C(M) = b(M) = \frac{d!}{(d - b + 1)! \prod_i m_i!}.$$

The input from [3] is geometric. It identifies $C(M)$ with an integral over the closure $C_{n,r}$ of the curvilinear locus of length- n subschemes of $\mathbb{C}^r$ ($n = d + 1$). It lowers r by a multiplicity-one hyperplane lemma while every exponent stays positive. On the plane sector this reaches $r = 2$, the punctual Hilbert scheme of the plane. There, via Lehn–Sorger, the integral becomes a coefficient in the class algebra:

$$C(M) = [\sigma] m_\nu(J_1, \dots, J_n), \quad J_k = \sum_{a < k} (a \ k),$$

where σ is any n -cycle and ν is the partition formed by the entries of $1 - M$. Everything below is new.

Lemma 3. *Let $\sigma(x) = x + 1 \pmod{n}$. A minimal monotone factorisation of σ is an expression $\sigma = (a_1 b_1)(a_2 b_2) \cdots (a_{n-1} b_{n-1})$ with $a_i < b_i$ and $b_1 \leq \cdots \leq b_{n-1}$. Its exponent vector is $e_k = \#\{i : b_i = k\}$, with $e_1 = 0$. Then for every $e \in \mathbb{Z}_{\geq 0}^n$ with $\sum e_k = n - 1$:*

$$\#\{\text{minimal monotone factorisations of } \sigma \text{ with exponent vector } e\} = \begin{cases} 1 & \text{if } D(j) := j - 1 - \sum_{k \leq j} e_k \geq 0 \text{ for all } j \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Necessity. The $n - 1$ transpositions of a minimal factorisation of an n -cycle are the edges of a tree on $[n]$. Those with $b_i \leq j$ are edges of a forest on $[j]$, so there are at most $j - 1$ of them.

Existence and uniqueness, by induction on n . The case $n = 1$ is trivial. Let $m = e_n$ and let $(a_1 n) \cdots (a_m n)$ be the final m factors, with product C . The remaining factors form a minimal monotone factorisation of $F = \sigma C^{-1}$, a permutation of $[n - 1]$ with exactly m cycles. Since $C^{-1} = (n \ a_1 \ a_2 \cdots a_m)$ we have $a_1 = \sigma^{-1}(n) = n - 1$ and

$$F(x) = x + 1 \ (x \notin \{a_i\}), \quad F(a_i) = a_{i+1} + 1 \ (i < m), \quad F(a_m) = 1.$$

Because $F \cdot C = \sigma$ is one cycle and C is an $(m + 1)$ -cycle, each cycle of F contains exactly one a_i . Starting at $a_{i+1} + 1$ and stepping by $+1$, the cycle through a_i must meet a_i before any other a_j . So $a_{i+1} < a_i$ with no a_j in between, and a_m is the least. Hence $a_1 > a_2 > \cdots > a_m$, and F is the product of the increasing cycles on the consecutive intervals

$$I_m = [1, a_m], \quad I_{m-1} = [a_m + 1, a_{m-1}], \quad \dots, \quad I_1 = [a_2 + 1, n - 1].$$

Transpositions on disjoint intervals commute, so monotone factorisations of F are exactly shuffles, merged by b , of monotone factorisations of the interval cycles. Minimality on each interval says the exponents inside I_{m-i} sum to $|I_{m-i}| - 1$. Since D changes by $1 - e_k \leq 1$ per step, this means $D(a_{m-i}) = i$, and the ballot condition inside the interval means $D > i$ strictly after a_{m-i} , up to the next cut. So a_{m-i} is the last j with $D(j) = i$. When e is ballot these last visits exist, are increasing in i , and end at $D(n - 1) = m - 1$. So the cut points exist and are forced, and the induction applies to each interval (each is again an increasing cycle, of smaller length). $\square$

Proof of Theorem 2. Expanding $m_\nu(J_1, \dots, J_n) = \sum_e \prod_k J_k^{e_k}$ over distinct rearrangements e of ν (padded by zeros, $e_1 = 0$ since $J_1 = 0$), the coefficient of σ counts pairs $(e, \text{minimal monotone factorisation with exponent vector } e)$. By Lemma 3 this is the number of ballot vectors e . Setting $\alpha_k = 1 - e_{k+1}$ ($k = 1, \dots, d$) turns the condition $D \geq 0$ into $\alpha_1 + \cdots + \alpha_j \geq 0$, and e into an ordering of M . So $C(M) = b(M)$. For the product formula, apply the cycle lemma to $-M$ with a final -1 appended: n integers ≥ -1 summing to -1 , exactly one cyclic rotation of which has all proper prefix sums nonnegative. Reversal identifies these rotations with ballot orderings of M , so $b(M) = \frac{1}{n} \cdot \frac{n!}{(d-b+1)! \prod_i m_i!}$. This is Kreweras' count [4]. $\square$

Lemma 3 is checked exhaustively for $n \leq 7$; the totals are the Catalan numbers 1, 2, 5, 14, 42, 132. The identity $C(M) = b(M)$ is checked on every plane-sector packet for $d \leq 7$ directly from the chamber series. Both checks are in tier 12 of the test suite.

Remark 4. On the plane sector the Morin chamber series is therefore indistinguishable, packet by packet, from the pure chain series $\prod_j (1 - x_j)^{-1}$. For example, at $d = 7$ the packet $M = (1, 1, 1, 0, -1, -1, -1)$ has 35 ballot orderings. Eight of them carry nonzero A_β , namely 1, 4, -1, -4, 5, 6, 6, 18, and these sum to 35.

3 The ballot conjecture

Conjecture 5 (ballot). *For every zero-sum multiset M of d integers, $C(M) \geq b(M)$, with equality if and only if $\max M \leq 1$. Equivalently: the series $F_d - \prod_j (1 - x_j)^{-1}$ has nonnegative packet sums, and they vanish exactly on the plane sector.*

Corollary 6. *Conjecture 5 implies Rimányi’s conjecture, and more: every Chern coefficient of Tp_{A_d} is at least 1 whenever the monomial occurs.*

This explains a feature of the data that had no explanation: across $d = 4, \dots, 7$ the least Chern coefficient is 1, never 0. It is attained exactly at $M = 0$ and $M = (1^{d-1}, 1 - d)$, the two multisets with $b(M) = 1$ on the plane sector.

Table 1: The ballot conjecture on every exact table. A violation is $C(M) < b(M)$; the equality columns test that $C = b$ holds exactly on the plane sector.

d	box	packets	violations	plane $C \neq b$	off-plane $C = b$	$\min_{\max M \geq 2} C/b$
2	level box L=14	15	0	0	0	2
3	level box L=14	169	0	0	0	2
4	level box L=12	1033	0	0	0	9/4
5	level box L=12	7166	0	0	0	12/5
6	level box L=11	29941	0	0	0	79/30
7	level box L=6	8033	0	0	0	58/21
7	charge box p=6	8033	0	0	0	58/21

Evidence. Table 1 tests the conjecture on every exact table available. The data come from the exact sweep of `chernpp.bboxes`, reconstructed by CRT and verified against a held-back prime. The level box $\min M \geq -L$ contains every packet of Tp^ℓ with $\ell < L$; the charge box contains every packet with $\max M \leq p$. There is no violation, no plane-sector packet with $C \neq b$, and no off-plane packet with $C = b$. Off the plane sector $C(M)/b(M) \geq 2$ throughout. The minimum is 2 at $(2, -1, -1)$ and grows slowly with d (9/4, 12/5, 79/30, 58/21), attained on the family $(2, 1^{d-4}, -1^{d-2})$ and its neighbours.

What does not work.

- *Factorial weights.* $C(p, -1^p) = p!$, which suggests weighting each ballot word by $\prod_{\alpha_i > 0} \alpha_i!$. That bound fails already at $d = 3$.
- *Fusion.* Replacing a 2 by two 1s and counting plane-sector ballot words of the longer multiset gives some families exactly. For example, $C(2, 0^{d-2}, -2) = \binom{d+1}{3} + \binom{d}{2}$, where both terms are plane-sector counts. But the residual is negative elsewhere.
- *Local step weights.* Keep weight 1 on steps ≤ 1 and give each step of size 2 a weight depending on local data: its height, together with the neighbouring steps, the number of remaining steps, or the next return. Fitted exactly on the packets with one 2, every such model that is consistent within a single d is inconsistent across d .

- *Formal Jucys–Murphy extension.* Theorem 2 reads $C(M) = \sum_{\alpha} [\sigma] \prod_k J_{k+1}^{1-\alpha_k}$ over ballot α . At charge 2 this needs J_k^{-1} , which exists for $k = 2, 3$ (their eigenvalues, the contents of boxes 2 and 3, are nonzero). Where it can be evaluated, it returns the wrong values, some of them fractional ($C(2, 0, -2) = 7$ against $23/2$). At $d = 4$ the needed inverses do not exist.
- *Individual coefficients.* No bound on single A_{β} can work: they are unbounded below (-3622 at $d = 7$), and the smallest symmetrisation that clears all negatives grows with d [5].

4 Zero insertion

Appending a 0 to M adds a singleton block and multiplies the Chern monomial by $c_{\ell+1}$. For a zero-free base M_0 write $C_z = C(M_0 \sqcup 0^z)$ and $b_z = b(M_0 \sqcup 0^z)$. On the plane sector b_z is a Kreweras number, a polynomial in z . For example $b(1, -1, 0^z) = \binom{z+2}{2} = 1 + 2z + \binom{z}{2}$.

Conjecture 7 (zero insertion). *For every zero-free M_0 , the sequences C_z and $C_z - b_z$ have nonnegative forward differences of every order at $z = 0$:*

$$C_z - b_z = \sum_{k \geq 0} N_k(M_0) \binom{z}{k}, \quad N_k(M_0) \geq 0.$$

This refines Conjecture 5, since every M is some $M_0 \sqcup 0^z$. It holds on all 4925 zero-free bases available (`results/zero_insertion.json`), for C and for $C - b$, at every order that fits in $d \leq 7$. Where the differences terminate within range, C_z is a polynomial:

$$C(2, -2, 0^z) = 2 + 5z + 4 \binom{z}{2} + \binom{z}{3}, \quad C(3, -3, 0^z) = 4 + 13z + 16 \binom{z}{2} + 9 \binom{z}{3} + 2 \binom{z}{4}.$$

The degree equals $\sum_{a>0} a + \#\{a < 0\}$ in all four certified cases. That is too few to conjecture.

5 Where a proof has to go

In the geometric model of [3], a packet of charge p reduces by the hyperplane lemma only to the curvilinear locus $C_{n,p+1}$ in $\mathbb{C}^{p+1}$. The plane case $p = 1$ is Theorem 2. Conjecture 5 therefore says:

the integral of $m_{\nu}(E)$ over the curvilinear locus in $\mathbb{C}^{p+1}$ is at least the count of ballot words, which is what the same integral gives in the plane, and exceeds it as soon as $p \geq 2$.

Two routes suggest themselves.

1. *A degeneration.* Degenerate $C_{n,p+1}$ to a cycle containing a planar component with multiplicity one, plus effective components on which $m_{\nu}(E)$ is nonnegative. The report of [3] records that effective-component positivity fails for individual degeneration components, so the decomposition would have to be chosen with care.
2. *A charge-2 Jucys–Murphy calculus.* Find the analogue, for the curvilinear component of the punctual Hilbert scheme of $\mathbb{C}^3$, of the Lehn–Sorger description. In such a calculus the excess $C(M) - b(M)$ should count a positive family of *non-monotone* or *branched* factorisations.

The exact tables (`results/morin_a{6,7}.json`, 20,725 coefficients) are available to test either route.

References

- [1] L. Bérczi, A. Szenes, Thom polynomials of Morin singularities, *Ann. of Math.* 175 (2012) 567–629.
- [2] R. Rimányi, Thom polynomials, symmetries and incidences of singularities, *Invent. Math.* 143 (2001) 499–521.
- [3] Rimányi positivity: findings summary (29 July 2026), unrefereed; Theorem 5.4 and Lemma 5.3. [papers/rimanyi_positivity_findings_summary.pdf](#).
- [4] G. Kreweras, Sur les partitions non croisées d’un cycle, *Discrete Math.* 1 (1972) 333–350.
- [5] Chern++, *Beyond Morin, and deeper into A_7* , [report/morin_d7.pdf](#).